

EPUSTAKA MUNSHI: SMART LIBRARY SYSTEM
WITH OCR LEDGER DIGITIZATION

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UNIVERSITI TEKNOLOGI MALAYSIA

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EPUSTAKA MUNSHI: SMART LIBRARY SYSTEM
WITH OCR LEDGER DIGITIZATION

WAN ZAFIRZAN BIN WAN TARMIZAN

A thesis submitted in fulfilment of the
requirements for the award of the degree of
Bachelor of Computer Science (Software Engineering)

Malaysia-Japan International Institute of Technology
Universiti Teknologi Malaysia

DECEMBER 2025

DECLARATION

I declare that this thesis entitled “*EPUSTAKA MUNSHI: SMART LIBRARY SYSTEM WITH OCR LEDGER DIGITIZATION*” is the result of my own research except as cited in the references. The thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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DEDICATION

This thesis is dedicated to my father and mother who raised me with nothing but kindness. Their unending support through hardship is a testament of a family's love in which I intend to honor by completing this thesis and to graduate with the best possible achievement that I can accomplish.

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ABSTRACT

The management of library resources in secondary schools is critical for fostering academic excellence. At SMK Abdullah Munshi, the library currently relies on traditional manual methods, specifically using handwritten physical ledgers to record book inventories and borrowing transactions. This manual approach has led to significant operational inefficiencies, including difficulty in tracking book status, deterioration of physical records, and a time-consuming borrowing process. To address these challenges, this project proposes the development of "ePustaka Munshi: Smart Library System with OCR Ledger Digitization." This web-based application is designed to modernize the library's workflow by integrating Optical Character Recognition (OCR) technology to digitize legacy handwritten records into a centralized database. Developed using the Agile Scrum methodology, the system utilizes React.js for the user interface and Supabase for backend data management. Key features include a specialized OCR verification module for librarians, a simplified barcode scanning interface for student assistants, and a role-based access system to ensure data security. The outcome of this project is a functional, scalable digital platform that preserves historical data while significantly reducing the administrative workload for library staff, ultimately creating a more efficient library environment for the school community.

ABSTRAK

Pengurusan sumber perpustakaan di sekolah menengah adalah kritikal dalam memupuk kecemerlangan akademik. Di SMK Abdullah Munshi, perpustakaan sekolah pada masa ini masih bergantung kepada kaedah manual tradisional, khususnya penggunaan buku rekod fizikal (lejar) tulisan tangan untuk merekod inventori buku dan transaksi peminjaman. Pendekatan manual ini telah menyebabkan ketidakcekapan operasi yang ketara, termasuk kesukaran menjejak status buku, kerosakan rekod fizikal, dan proses peminjaman yang memakan masa. Bagi menangani cabaran ini, projek ini mencadangkan pembangunan "ePustaka Munshi: Sistem Perpustakaan Pintar dengan Pendigitalan Lejar OCR." Aplikasi berasaskan web ini direka untuk memodenkan aliran kerja perpustakaan dengan mengintegrasikan teknologi Pengesanan Aksara Optik (OCR) bagi mendigitalkan rekod lama tulisan tangan ke dalam pangkalan data berpusat. Dibangunkan menggunakan metodologi Agile Scrum, sistem ini menggunakan React.js untuk antara muka pengguna dan Supabase untuk pengurusan data bahagian belakang. Ciri-ciri utama termasuk modul pengesahan OCR khusus untuk pustakawan, antara muka imbasan kod bar yang ringkas untuk pembantu pusat sumber, dan sistem capaian berasaskan peranan untuk memastikan keselamatan data. Hasil projek ini ialah sebuah platform digital berfungsi dan berskala yang memelihara data sejarah sambil mengurangkan beban kerja pentadbiran kakitangan perpustakaan secara drastik, sekaligus mewujudkan persekitaran perpustakaan yang lebih cekap untuk warga sekolah.

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LIST OF ABBREVIATIONS

ANN	-	Artificial Neural Network
GA	-	Genetic Algorithm
PSO	-	Particle Swarm Optimization
MTS	-	Mahalanobis Taguchi System
MD	-	Mahalanobis Distance
TM	-	Taguchi Method
UTM	-	Universiti Teknologi Malaysia
XML	-	Extensible Markup Language
ANN	-	Artificial Neural Network
GA	-	Genetic Algorithm
PSO	-	Particle Swarm Optimization

LIST OF SYMBOLS

δ	-	Minimal error
D, d	-	Diameter
F	-	Force
v	-	Velocity
p	-	Pressure
I	-	Moment of Inertia
r	-	Radius
Re	-	Reynold Number

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The effective management of library resources is essential in supporting the academic growth of students and maintaining organized access to information. At SMK Abdullah Munshi, the library plays a significant role in cultivating reading habits and academic excellence, especially among secondary school students in Penang, Malaysia. However, much of the library's operations still depend on traditional methods, including handwritten ledgers, manual transaction logging, and basic spreadsheet listings that lack validation and real-time updates. These limitations highlight the need for a modernized digital solution that can enhance efficiency, accuracy, and accessibility.

To address these needs, this project introduces a web-based Smart Library Management System equipped with Optical Character Recognition (OCR) to automatically digitize legacy handwritten records. By integrating OCR, barcode scanning, and role-based user modules, the system aims to transform manual library workflows into a centralized digital platform. This modernization not only preserves historical book data but also reduces librarian workload and improves student participation in library activities.



Figure 1.1 SMK Abdullah Munshi Logo

1.2 Problem Background

SMK Abdullah Munshi's library currently manages over 7,000 books using a physical ledger that has been manually updated for many years. Important information such as book titles, authors, publishers, and Dewey classifications are handwritten, making the records vulnerable to human error, inconsistent formatting, and physical deterioration. Although an Excel file exists as a partial digital record, it lacks database functions such as duplication checks, borrowing status tracking, and data validation, resulting in inaccurate or incomplete information.

Additionally, the borrowing and returning of books are entirely manual processes where librarians record each transaction by hand. This process is slow, prone to oversight, and makes it difficult to determine which books are currently loaned out. The school previously used a digital system called e-Kutub Khanah, but due to technical and maintenance challenges, it is no longer operational, forcing the library to revert to fully manual procedures.

Students and library assistants often help with book cataloguing, but without a structured system, the work becomes tedious and inconsistent. The lack of a

streamlined digital workflow reduces efficiency, increases the likelihood of lost or duplicated records, and discourages students from participating actively. These ongoing issues highlight the need for a modern library management solution that automates data entry, digitizes historical records, and simplifies daily operations for both library staff and students.

1.3 Project Aim

The aim of this project is to develop a web-based Smart Library Management System that digitalizes handwritten records using OCR and improves daily library operations at SMK Abdullah Munshi.

1.4 Project Objectives

The objectives of the project are:

- (a) To analyze and design a library management system that addresses issues in manual record-keeping, book cataloguing, and borrowing activities.
- (b) To develop a web-based system equipped with OCR to digitize handwritten ledger entries and store them in a centralized database.
- (c) To implement barcode-based borrowing and returning modules along with role-based access for librarians and student assistants.
- (d) To evaluate the system's functionality, accuracy, and usability through testing and feedback from library staff and students.

1.5 Project Scope

The scopes of the project are:

- (e) The system will include modules for book management, student records, borrowing and returning transactions, OCR digitization, and reporting.
- (f) The system will be accessible through web browsers on school computers, using Supabase as the backend and React for the interface.
- (g) OCR will be used specifically for digitizing existing handwritten book records from the ledger.
- (h) Features such as mobile applications, online fine payments, and integrations with external databases are excluded from the scope.

1.6 Project Importance

This project is important because it modernizes the library operations at SMK Abdullah Munshi, which currently relies heavily on manual record-keeping. By introducing OCR to digitize handwritten ledger entries, the system preserves historical data that has been maintained for years. The digital catalog also reduces errors, speeds up book searches, and helps librarians manage records more efficiently. Barcode-based borrowing further supports a smoother workflow and minimizes mistakes in tracking borrowed books. Overall, the system aims to improve accuracy, reduce workload, and create a more organized and accessible library environment for both staff and students.

1.7 Report Organization

This report is structured to provide a clear explanation of how the proposed system was developed and evaluated. Chapter 2 covers the literature review, which includes the case study, analysis of the current system, comparison with existing solutions, and a discussion of the technologies involved. Chapter 3 outlines the system development methodology, detailing the chosen approach, its justification, the development phases, technologies used, and the system requirements. Chapter 4 presents the requirement analysis and design, including project requirements, system architecture, database design, and interface design. Chapter 5 describes the implementation and testing phases, highlighting the development of core functions, interface construction, and system testing. Lastly, Chapter 6 summarizes the project outcomes and provides recommendations for future enhancements.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

2.1 Introduction

This chapter reviews relevant background information, existing practices, and technologies related to the development of a Smart Library Management System. It begins with an inter-organisation case study of SMK Abdullah Munshi, serving as the direct stakeholder for this project. The chapter then examines the current library system used by the school, followed by a comparison of existing library management systems commonly adopted in Malaysian educational institutions. Next, the technologies applied in the proposed system, such as Optical Character Recognition (OCR), barcode scanning, and web-based database systems, are reviewed. The chapter concludes with a summary highlighting key findings that justify the proposed solution.

2.2 Case Study

SMK Abdullah Munshi is a secondary school located in Penang, Malaysia, with an active school library that supports teaching and learning activities for students and teachers. The school library manages a collection of more than 7,000 physical books accumulated over several decades. The library is operated by a librarian with assistance from student library helpers.

This project uses SMK Abdullah Munshi as a direct stakeholder case study. Information about the current library operations was obtained through continuous communication with the school librarian, Cikgu Farah, via WhatsApp messages and

online meetings. The librarian also shared photographs of the library environment, including the physical ledger used for book records and the book processing area. These materials provided valuable insight into the daily workflow, challenges, and limitations faced by the library.

Based on the case study, it was observed that most library processes remain manual. Book information is recorded in a large handwritten ledger, and borrowing activities are logged manually. Although a basic digital book list exists in spreadsheet form, it functions only as a static record and does not support automation or validation. These findings highlight the need for a practical and low-cost digital solution that suits the school's operational and financial constraints.

2.3 Current System Analysis

The current library system at SMK Abdullah Munshi relies heavily on manual documentation. All book records, including title, author, publisher, and Dewey classification, are written by hand in a physical ledger book. This ledger has been maintained for many years and now contains thousands of entries. While the ledger preserves historical data, it is difficult to search, update, or verify information efficiently.

Book borrowing and returning are also handled manually. Librarians record the borrower's name, book title, and return date by writing in a logbook. This process is time-consuming and prone to errors such as missing entries, unclear handwriting, and incorrect return dates. It is also difficult to quickly determine whether a book is currently available or borrowed.

Students and library assistants often help with cataloguing new books. However, without a structured data entry system, book details may be entered inconsistently, leading to duplicate or incomplete records. These manual processes increase the librarian's workload and reduce overall efficiency. The limitations of the current system demonstrate the need for a centralized, searchable, and automated library management solution.

2.4 Comparison between existing systems

System / Method	Key Features	Strengths	Limitations
Manual Ledger & Excel Spreadsheet	Handwritten book records, basic Excel list	Low cost, simple to use, preserves historical records	Time-consuming, error-prone, no search automation, no borrowing status tracking, no data validation
KOHA Library System	Cataloguing, circulation, reporting, user management	Open-source, feature-rich, widely used in large libraries	Requires server setup and technical expertise, complex interface, unsuitable for small school libraries
e-Kutub Khanah	Book cataloguing, barcode scanning for borrowing	Previously supported scanning and automation	No longer actively maintained, technical issues, limited support
Advance Integrated NILAM System (AINS)	Student reading (NILAM) tracking, online reporting	Supports NILAM programme, centralized reading records	Does not manage physical book inventory or borrowing transactions

Proposed Smart Library Management System	OCR ledger digitization, book management, barcode-based borrowing, role-based access	Designed for school environment, low-cost, easy to use, preserves handwritten records, minimal maintenance	Limited to core library functions, no online payment or AI-based recommendations
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2.5 Literature Review of Technology Used

Optical Character Recognition (OCR)

OCR technology is used to convert scanned images of handwritten or printed text into machine-readable data. In this project, OCR is applied to digitize handwritten entries from the library ledger. Librarians can review and correct extracted text before saving it into the database, ensuring data accuracy while preserving historical records.

Barcode and QR Code Technology

Barcode or QR code scanning is used to automate borrowing and returning processes. Each book is assigned a unique code that can be scanned using a low-cost USB scanner connected to a computer. This approach reduces manual writing and allows real-time tracking of book availability.

Web-Based Application

The system is developed as a web-based application to ensure accessibility from school computers without requiring mobile devices. A browser-based system reduces installation complexity and suits the school's existing infrastructure.

Database System

A free and locally hostable database system such as PostgreSQL or MySQL is proposed. These databases are open-source, reliable, and suitable for small-scale deployment. They can be hosted on a local school computer or server without subscription costs, making them practical for schools with budget limitations.

Role-Based Access Control

Role-based access ensures that librarians, student assistants, and general users have appropriate permissions. This improves data security and accountability while allowing student involvement under controlled access.

2.6 Chapter Summary

This chapter discussed the background and context of the proposed system through an inter-organisation case study of SMK Abdullah Munshi. The current manual library system and its limitations were analyzed, followed by a comparison with existing library management systems. Relevant technologies such as OCR, barcode scanning, web-based systems, and open-source databases were reviewed. The findings from this chapter provide a strong justification for developing a Smart Library Management System that addresses the specific needs and constraints of a Malaysian secondary school library.

CHAPTER 3

SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

This chapter discusses the methodology adopted for the development of the ePustaka Munshi: Smart Library System. The methodology serves as a framework to structure, plan, and control the process of developing the information system. For this project, the Agile Development Methodology was selected.

This approach was chosen because it emphasizes iterative development, allowing the system to be built in small, manageable increments known as "Sprints". Given the complexity of integrating Optical Character Recognition (OCR) for digitizing handwritten ledgers, an iterative approach is crucial. It allows for continuous testing and refinement of the OCR accuracy based on real-world feedback from the stakeholder, the school librarian.

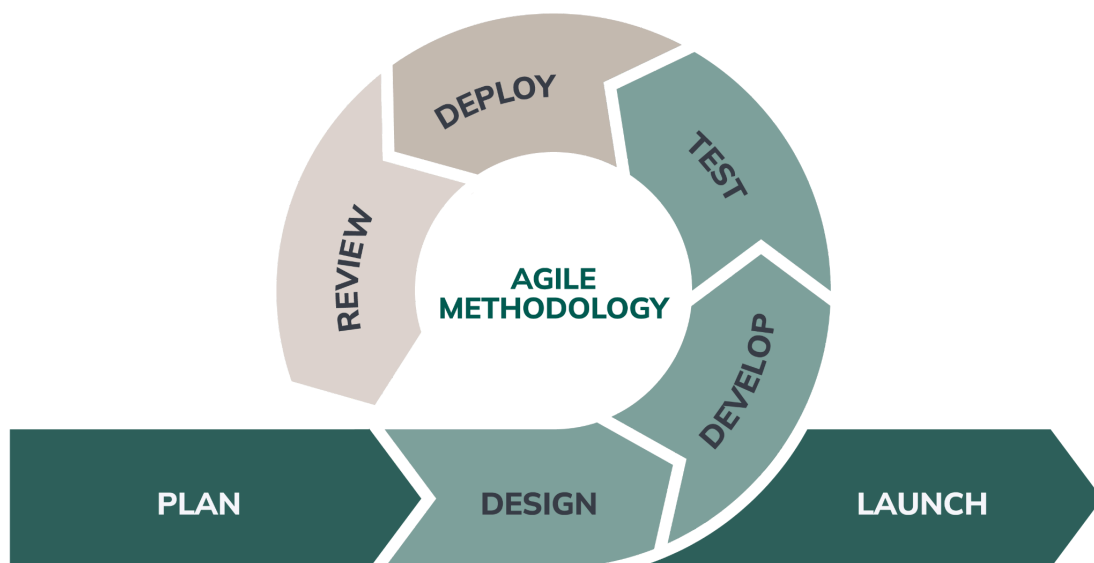


Figure 3.1: Agile Development Lifecycle Diagram

3.2 Methodology Choice and Justification

The Agile Model was chosen over traditional models like Waterfall for several specific reasons related to the constraints and requirements of SMK Abdullah Munshi:

- **Iterative Refinement of OCR Features:** The project involves digitizing legacy handwritten records, which varies significantly in handwriting quality. Agile allows the developer to build a basic OCR prototype, test it with actual ledger images, and refine the algorithm in subsequent iterations without halting the entire project.
- **Stakeholder Involvement:** The librarian, Cikgu Farah, is a key stakeholder who provides ongoing feedback. Agile encourages regular reviews (Sprint Reviews), ensuring that features like the "Barcode Borrowing System" meet the practical needs of the library staff before the final delivery.
- **Flexibility to Change:** Requirements in school environments often evolve. For instance, the need to separate access rights between "Student Assistants" and "Librarians" might require adjustments to the database schema mid-development. Agile accommodates these changes more effectively than rigid models.

3.2.1 Sprint 1: User & Role Management

This initial sprint establishes the core security and access infrastructure. It focuses on the User & Role Management Module.

- **Objective:** To enable secure access for Librarians, Student Assistants, and Students.
- **Key Deliverables:**
 - **User Login (US001):** Implementation of secure login to direct users to their specific dashboards.

- **Student Account Management (US002):** Capability for Librarians to manage student data.
- **Assistant Account Management (US003):** Creation of accounts for student assistants who help with daily operations.

3.2.2 Sprint 2: Book Catalog & Inventory

Sprint 2 focuses on the Book Catalog & Inventory Module. This replaces the static Excel sheets with a dynamic database.

- **Objective:** To provide a searchable, digital inventory of all library assets.
- **Key Deliverables:**
 - **View Catalog (US004):** A public-facing interface for students to browse books.
 - **Search & Filter (US005):** Search functionality to find books by title, author, or category.
 - **Manage Book Records (US006):** CRUD capabilities for Librarians to manually add or edit book details.

3.2.3 Sprint 3: Ledger Digitization & OCR Integration

This sprint addresses the project's unique value proposition the Ledger Digitization & OCR Module. This is the most technically complex phase.

- **Objective:** To automate the migration of historical handwritten data into the system.
- **Key Deliverables:**
 - **Upload Ledger Image (US007):** Interface to upload high-quality images of physical logbooks.
 - **Verify OCR Data (US008):** A review interface where Librarians can correct text extracted by the OCR engine before it is committed to the database.
 - **Save Digitized Records (US009):** The final commitment of verified data into the main Book Catalog.

3.2.4 Sprint 4: Borrowing & Returning Management

The final sprint focuses on the Borrowing & Returning Management Module, integrating hardware (barcode scanners) with the software.

- **Objective:** To streamline daily circulation tasks and maintain accurate book status.
- **Key Deliverables:**
 - **Scan for Borrowing (US010):** Real-time status updates when a book is checked out.
 - **Scan for Returning (US011):** automated return processing to make books available again.
 - **Transaction Recording (US012 & US013):** Logging the borrower's history and updating inventory status automatically.

3.3 Phases of the Chosen Methodology

The Agile Scrum methodology divides the development process into iterative cycles. The following phases outline how the methodology is applied to the ePustaka Munshi project:

1. **Product Backlog Creation:** The backlog was created based on the requirements elicited from the SRS. It contains all User Stories (US001 - US013) prioritized by dependency (e.g., User Accounts must exist before Borrowing can occur).
2. **Sprint Planning:** At the start of each sprint, specific user stories are selected. For example, Sprint 3 is dedicated entirely to OCR tasks to ensure focus on the complex image processing logic.
3. **Sprint Execution & Standups:** Development of the selected features (coding, database integration). Daily self-reviews are conducted to ensure the OCR engine integration is progressing without blocking the main application.
4. **Sprint Review & Testing:** At the end of each sprint, the features are tested against the Acceptance Criteria defined in the SRS (e.g., "System must respond to login within 3 seconds").

3.4 Technology Used Description

The technology stack was selected to ensure the system is web-based, cost-effective, and compatible with the school's existing Windows-based hardware.

Category	Technology	Purpose	Justification
Frontend	React.js	User Interface Development	Provides a responsive, component-based UI that is essential for the interactive OCR verification screen ²⁸ .
Backend / DB	Supabase	Database & Auth	Offers a scalable PostgreSQL database to store book records and user data without complex server maintenance.
OCR Engine	Tesseract.js / Vision API	Optical Character Recognition	Required to convert the handwritten text from the uploaded ledger images into digital strings ²⁹ .
Hardware	Barcode Scanner	Input Device	USB-based scanners act as keyboard inputs to rapidly enter book IDs during borrowing ³⁰ .
Design	UML	Modeling	Used for creating Use Case, Activity, and

			Sequence diagrams to visualize system flow.
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Table 3.4.1 Technology Stack Table

3.5 System Requirement Analysis

This project focuses on developing a web application accessible via standard web browsers. The requirements are categorized into Included Features, Excluded Features, and Functional/Non-Functional Requirements based on the SRS.

Feature	Description
OCR Ledger Digitization	Capability to upload images of handwritten ledgers and extract book data for digital storage ³³ .
Barcode Transaction	Automating the borrowing and returning process using USB barcode scanners ³⁴ .
Role-Based Access	Distinct interfaces and permissions for Librarians (Admin), Student Assistants, and Students ³⁵ .
Centralized Database	A unified repository for Book, Student, and Transaction data to replace manual logbooks.

Table 3.5.1 Features Table

Feature	Description
Mobile Application	No native iOS or Android app will be developed; the system is strictly web-based.
Online Payments	The system will not handle fines or monetary transactions.
External Integration	No integration with National Library systems or third-party digital libraries.

Table 3.5.2 Excluded Features Table

Category	Requirement
Authentication	Users must log in with valid credentials to access the system (US001).
Inventory	The system allows adding, editing, and deleting book records (US006).
Digitization	The system must accept image uploads and present OCR data for verification (US008).
Circulation	The system must validate book availability before recording a borrow transaction (US012).

Table 3.5.3 Functional Requirements Table

Category	Requirement
Performance	The system shall respond to user login requests within 3 seconds.
Throughput	The system shall support at least 50 concurrent users performing standard operations.
Availability	The system shall be available 99% of the time excluding maintenance.
Reliability	The system must ensure that incorrect OCR data is verified before saving.

Table 3.5.4 Non-Functional Requirements Table

3.6 Chapter Summary

This chapter detailed the Agile Scrum methodology used to develop the ePustaka Munshi system. The iterative nature of Agile is essential for managing the complexity of the OCR integration and ensuring the system meets the practical needs of the school. The development was structured into four specific sprints: User Management, Book Cataloging, Ledger Digitization (OCR), and Transaction Management. The technology stack, including React and Supabase, was selected to provide a cost-effective and scalable solution. The next chapter will detail the specific analysis and design of these requirements.

CHAPTER 4

REQUIREMENT ANALYSIS AND DESIGN

4.1 Introduction

This chapter presents the requirement analysis and system design for the proposed Smart Library Management System developed for SMK Abdullah Munshi. It serves as a detailed blueprint that translates the issues identified in the current manual library operations into structured system requirements and design specifications. The discussion in this chapter builds upon the problem background and project objectives outlined in Chapter 1.

The requirement analysis focuses on understanding the functional and non-functional needs of the system through direct engagement with the library stakeholder, including interviews, online discussions, and observations of existing practices such as handwritten ledger usage and manual borrowing procedures. These activities help ensure that the system requirements accurately reflect the operational environment of the school library.

In addition, this chapter outlines the overall system design, including the architectural approach, module organization, and interaction between system components. The database design is presented to illustrate how library data such as books, users, and borrowing transactions are structured and managed. Interface design considerations are also discussed to ensure usability for librarians, student assistants, and students.

Overall, this chapter provides a comprehensive explanation of how requirements were gathered and transformed into a coherent system design, forming the foundation for the implementation and testing phases described in the subsequent chapters.

4.2 Requirement Analysis

Cikgu Farah, the librarian of SMK Abdullah Munshi, is the primary stakeholder involved in this system. A series of informal interviews and discussions were conducted with her to identify the operational challenges faced in managing the school library. These interactions were carried out through WhatsApp messages and Google Meet sessions, allowing continuous communication throughout the requirement analysis phase. Through these discussions, several key issues were identified, particularly those related to the heavy reliance on handwritten records and manual borrowing procedures.

From the stakeholder feedback, it was observed that the existing library system depends on a physical ledger that has been used for many years to record book information such as titles, authors, publishers, and classification numbers. This manual process requires significant effort to maintain and is prone to inconsistencies, missing data, and human error. In addition, the manual recording of borrowing and returning activities makes it difficult to track book availability and identify overdue items efficiently. These issues were highlighted as major operational burdens for the librarian and were therefore selected as the primary focus of this Final Year Project due to their direct impact on daily library operations.

In addition to verbal discussions, supporting materials were provided by the stakeholder to facilitate a clearer understanding of the current workflow. These included photographs of the library space, the handwritten ledger book, and the book storage arrangement. These materials helped in visualizing the existing processes and identifying practical constraints such as limited hardware resources and the absence of mobile devices for library use. The information gathered from these materials was used to determine realistic system requirements that align with the school's operational environment.

Beyond scheduled discussions, continuous clarification was achieved through informal communication with the stakeholder. This ongoing interaction was essential for validating functional requirements such as role-based access, barcode-based

borrowing, and the scope of OCR usage for digitizing historical records. Feedback obtained during this phase also helped refine system boundaries to ensure the proposed solution remained feasible within the time and resource limitations of the project.

Based on the collected information and analysis, a Software Requirements Specification (SRS) document was produced to formally capture the functional and non-functional requirements of the system. This document serves as the baseline reference for system design, implementation, and testing. For further details, the complete Software Requirements Specification (SRS) is provided in Appendix B, while supporting materials such as stakeholder communication records and reference images are included in the appendices where applicable.

4.2.1 Functional Requirement

The functional requirements describe the core capabilities of the Smart Library Management System and define how the system supports daily library operations at SMK Abdullah Munshi. These requirements are designed to address the inefficiencies of the existing manual processes while ensuring the system remains practical for the school's operational environment. The system functionalities are organized based on the different user roles interacting with the system, namely the Librarian, Student Assistant, and Student (Borrower).

Each user role is provided with specific functions aligned with their responsibilities. The overall functional requirements are derived from stakeholder discussions, observation of the current library workflow, and the project objectives outlined in Chapter 1.

Use case	Function	Description
UC001	Login to System	This use case allows authorized users (Librarian and Student Assistant) to log in to the system using valid credentials in order to access system functionalities based on their roles.
UC002	Manage Student Accounts	This use case allows the Librarian to create, update, view, and deactivate student user accounts to control access to the library system.
UC003	Manage Student Assistant Accounts	This use case allows the Librarian to manage Student Assistant accounts, including creating, updating, and assigning permissions for assisting library operations.
UC004	View Book Catalog	This use case allows users to view the list of available books in the library, including basic details such as title, author, and availability status.
UC005	Search and Filter Books	This use case allows users to search for books and apply filters (e.g., title, author, category, availability) to quickly locate specific books in the catalog.
UC006	Manage Book Records	This use case allows the Librarian or Student Assistant to add new books, edit existing book details, or delete book records from the system database.
UC007	Upload Ledger Image for OCR	This use case allows the Librarian or Student Assistant to upload scanned images of physical book ledgers for Optical Character Recognition (OCR) processing.
UC008	Verify and Correct OCR Data	This use case allows users to review, validate, and manually correct extracted OCR data before it is saved into the system to ensure data accuracy.
UC009	Save Digitized Records	This use case allows the system to store verified and corrected digitized book records into the database for future access and management.

UC010	Scan Book for Borrowing	This use case allows the Librarian or Student Assistant to scan a book barcode to initiate a borrowing transaction for a student.
UC011	Scan Book for Returning	This use case allows the Librarian or Student Assistant to scan a book barcode to process the return of a borrowed book.
UC012	Validate Book Availability	This use case allows the system to verify whether a book is available for borrowing or already on loan before completing a transaction.
UC013	Update Book Status	This use case allows the system to update the book's availability status and borrowing records after a borrow or return transaction is completed.

**Table 4.2.1 *Description of Module and Functions for ePustaka Munshi:
Smart Library System***

Librarian Functions

The librarian serves as the primary authority within the system and is responsible for overseeing library operations. The system allows the librarian to log in securely to access all administrative functions (UC001). Once authenticated, the librarian can manage student and student assistant accounts, including creating, updating, or deactivating user records to maintain controlled access to the system (UC002, UC003).

Book management is a core responsibility of the librarian. The system enables the librarian to add new book records, edit existing information, and remove outdated or damaged items from the catalog (UC006). To support the transition from manual records, the librarian can upload images of handwritten ledger pages, which are processed using Optical Character Recognition (OCR) technology (UC007). Extracted data can then be reviewed and corrected before being saved into the database to ensure accuracy and consistency (UC008, UC009).

In addition, the librarian can monitor borrowing activities, identify overdue books, and view borrowing histories to support administrative decisions and disciplinary actions when necessary.

Student Assistant Functions

Student assistants support the librarian in daily library operations. After logging into the system, they are granted limited but essential access to assist with circulation tasks. Student assistants can scan books using a barcode scanner to process

borrowing transactions, allowing the system to automatically record the borrower, due date, and transaction details (UC010).

They are also able to scan books during the return process, which updates the book's availability status in real time and closes the borrowing record accordingly (UC011). Through these functions, student assistants help reduce the librarian's workload while ensuring that borrowing and returning processes remain accurate and efficient.

Student assistants can also view book availability, borrowing status, and due dates to assist students and librarians in tracking outstanding loans.

Student (Borrower) Functions

Students interact with the system primarily as borrowers. After logging in, students can view the complete book catalog and browse available resources (UC004). To improve usability, the system provides search and filtering features that allow students to locate books based on criteria such as title, author, or category (UC005).

Students can view their personal borrowing history, current borrowed items, and due dates. The system also supports future scalability, such as online book reservations, favourites, and notification alerts when due dates are approaching or when reserved books become available.

System-Driven Functions

Several system functions operate automatically to support user activities. These include validating user credentials during login, processing OCR results, updating book availability after borrowing or returning, and storing all transaction data in a centralized database. These automated processes ensure data consistency, reduce manual errors, and improve overall operational efficiency.

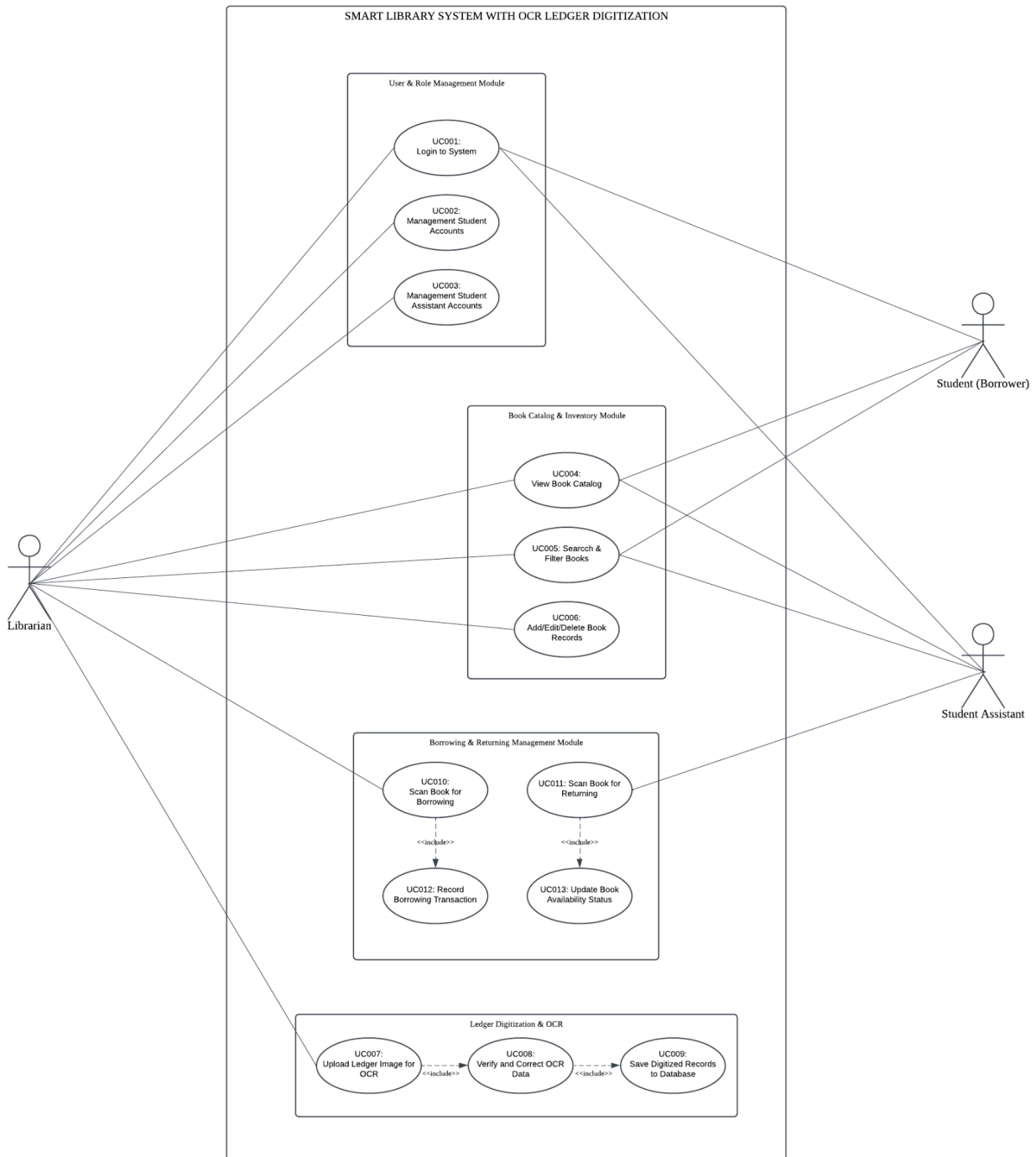


Figure 4.2.1: Use Case Diagram for EPUSTAKA MUNSHI: SMART LIBRARY SYSTEM WITH OCR LEDGER DIGITIZATION

4.3 Project Design

This section translates the requirements into a technical structure. The ePustaka Munshi system utilizes a modern web application architecture designed for scalability and maintainability.

4.3.1 System Architecture

The system follows a client-server architecture, leveraging cloud-based technologies to minimize the need for complex on-site server maintenance at the school.

1. **Presentation Layer (Frontend):** Developed using React.js, this layer is responsible for the user interface. It runs in the user's web browser and handles user interactions, such as displaying the book catalog, capturing barcode inputs, and rendering the image cropping tool for the OCR module.
2. **Application Logic & Database Layer (Backend):** The system utilizes Supabase, which acts as a Backend-as-a-Service (BaaS). Supabase manages the PostgreSQL database, authentication, and API endpoints. This approach eliminates the need to build a custom backend server from scratch, allowing for faster development and robust security.
3. **External Services (OCR Engine):** For the digitization feature, the system integrates the Tesseract.js engine (or Vision API). When a librarian uploads a ledger image, the frontend sends the image to this engine, which processes the handwriting and returns raw text strings back to the interface for verification.

4.4 Database Design

The database is the backbone of the system, replacing the physical logbooks previously used by the school. The design uses a relational database model (PostgreSQL via Supabase) to ensure data integrity and efficient querying.

Based on the domain analysis, the database schema is organized into the following key entities:

1. **Users:** Stores login credentials and profile information. This table includes a Role attribute to distinguish between Librarians, Assistants, and Students.
2. **Books:** This is the central repository for library assets. It stores details such as Title, Author, Publisher, ISBN, and crucially, the Current_Status (Available, Borrowed, Lost).
3. **Transactions:** To replace the manual borrowing logbook, this entity records every borrowing event. It links a Student_ID to a Book_ID and records the Borrow_Date, Due_Date, and Return_Date.
4. **OCR_Records:** This table is unique to this project. It stores the temporary data extracted from ledger images, including the raw text and the link to the uploaded image file, waiting for verification before being merged into the main *Books* table.

4.5 Interface Design

The user interface (UI) design focuses on clarity and workflow efficiency. The design process involved creating wireframes to visualize the flow before implementation.

4.5.1 Login and Dashboard

The entry point of the system is a secure login screen. Upon successful authentication, the system redirects the user to a role-specific dashboard.

- The **Librarian Dashboard** provides a high-level view of library statistics (e.g., total books, overdue items) and quick access to the OCR upload module.
- The **Student Assistant Dashboard** is streamlined, featuring a prominent "Scan to Borrow" and "Scan to Return" button to facilitate quick counter service.

4.5.2 OCR Digitization Interface

This is the most complex interface in the system. It is designed as a split-screen view. On one side, the uploaded image of the handwritten ledger is displayed. On the other

side, a form is pre-filled with the text extracted by the OCR engine. This layout allows the librarian to visually compare the handwriting against the digital text and correct any errors immediately before saving.

4.5.3 Borrowing and Returning Interface

To support the barcode scanner hardware, this interface keeps the input field in focus automatically. When a barcode is scanned, the system instantly fetches the book details and displays them for confirmation. A visual indicator (Green for Success, Red for Error) provides immediate feedback to the Student Assistant regarding the transaction status.

4.6 Test Case Design

To ensure a stable application, the test case design is derived directly from the functional requirements. Each test case is designed to verify that the system behaves as expected under specific conditions, including normal workflows and exception handling.

Table 4.1 below demonstrates a critical test case for the unique feature of this project: verifying and correcting OCR data before saving it to the database.

Test Case ID	TC008_01	Test Case Description	Verifies that a Librarian can manually correct errors in the text extracted by the OCR engine before saving to the database.	
Created by:	Wan Zafirzan Bin Wan Tarmizan	Version:	1.0	
No	Prerequisites		No.	Test Data
1	Librarian is logged in.		1	Image: ledger_page_5.jpg (Contains handwritten text: "Sejarah Melayu")

2	A ledger image has been uploaded and processed,		2	OCR Output (Simulated Error): "Sej@rah M3layu"
			3	Corrected Input: "Sejarah Melayu"
Test Conditions	Verify data correction workflow for imperfect OCR results.			
Step #	Step Details	Expected Result		
1	Navigate to "OCR Verification" screen.	The uploaded image is displayed on the left; the OCR extracted text is displayed in editable fields on the right.		
2	Locate the field containing "Sej@rah M3layu".	The field is editable and focuses when clicked.		
3	Manually type "Sejarah Melayu" to replace the error.	The field updates to reflect the new user input.		
4	Click "Save to Catalog".	1. System displays "Success" message. 2. The record is saved in the Books database table. 3. The saved title is		

		"Sejarah Melayu", NOT the original OCR error.	
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4.7 Chapter Summary

This chapter has detailed the analysis and design necessary to transform the manual library operations of SMK Abdullah Munshi into a digital ecosystem. By establishing clear functional and non-functional requirements, the boundaries of the system were defined. The architectural design leverages React and Supabase to provide a modern, scalable solution, while the database design ensures data integrity that was previously lacking in the manual ledgers. Finally, the interface design prioritizes the specific workflows of the school librarians and student assistants, ensuring the system is not only functional but also usable. These designs serve as the blueprint for the implementation and testing phases discussed in the next chapter.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 Introduction

This chapter details the transformation of the system design and requirements into a functional software product. It outlines the development environment, the coding implementation of core modules, and the testing strategies employed to validate the ePustaka Munshi system.

The implementation phase involved integrating three distinct technologies, the React.js frontend for user interaction, the Supabase backend for data management, and the Tesseract.js engine for the OCR capabilities. This chapter also discusses the challenges faced during the coding process, particularly regarding the accuracy of handwritten text recognition, and how these were addressed through interface design.

5.2 Coding of System Main Functions

The system development was executed based on the Agile sprints defined in the methodology. The coding process was divided into the frontend presentation layer and the backend logic layer.

5.2.1 Frontend Implementation (React.js)

The user interface was built using the React library to ensure a responsive experience on the school's desktop computers. The application state, such as the currently logged-in user (Librarian vs. Student Assistant), is managed using React Context to protect administrative routes. Key components developed include:

- **Dynamic Forms:** Reusable form components were created for book entry to ensure data validation (e.g., ensuring ISBNs are numeric) before submission.

- **OCR Processor:** A dedicated component was built to handle image file uploads. This component integrates the Tesseract.js library, which runs asynchronously in the browser to process images without freezing the user interface.

5.2.2 Backend Implementation (Supabase)

Supabase was chosen as the backend-as-a-service to accelerate development. The implementation involved:

- **Database Creation:** Tables for users, books, transactions, and ocr_logs were created using SQL within the Supabase dashboard.
- **Row Level Security (RLS):** Security policies were coded to enforce role-based access. For example, a policy was written to allow 'Student Assistants' to INSERT into the transactions table but block them from DELETE operations on the books table.

5.3 Interfaces of System Main Functions

This section highlights the actual user interfaces developed during the project, demonstrating how the requirements were translated into visual screens.

5.3.1 Login and Role-Based Dashboards

The login screen serves as the gateway to the system. Upon entry, the system logic checks the user's role in the database.

- **Librarian View:** Accesses the full suite of tools, including the "Digitize Ledger" and "Manage Users" buttons.
- **Student Assistant View:** A simplified interface containing only large, accessible buttons for "Borrow" and "Return" to minimize errors during busy break times.

5.3.2 OCR Digitization Workspace

This interface is the core innovation of the project. To handle the unpredictability of handwriting in the school's old ledgers, a split-pane layout was implemented. The left side displays the uploaded image of the physical ledger, allowing the librarian to

zoom and pan. The right side displays the data extracted by the OCR engine in input fields. This design forces a "human-in-the-loop" workflow, ensuring that the librarian verifies the data before it is permanently committed to the digital catalog.

5.3.3 Barcode Transaction Interface

Designed for speed, this screen keeps the focus on the barcode input field. When a Student Assistant uses the USB scanner, the input is captured immediately. The interface then dynamically fetches the student's name and book title from the database, displaying them on a card for visual confirmation before the transaction is finalized.

5.4 Testing

Testing was conducted continuously throughout the development lifecycle to ensure system reliability and usability within the school environment.

5.4.1 Black Box Testing

Black box testing focused on the functionality of the system without inspecting the internal code structure.

5.4.1.1 System Flow and Input Verification

Test cases were executed to validate the complete workflow of a library book.

- **Digitization Flow:** Tested uploading various image qualities (clear vs. blurry) to ensure the system handles OCR failures gracefully by allowing manual entry.
- **Borrowing Flow:** Tested scanning a book that is *already* borrowed. The system successfully prevented the transaction and displayed an error message: "Book is currently unavailable."

5.4.1.2 Error Handling

The system was tested against invalid inputs. For example, entering a non-existent Student ID during a borrowing session triggers a "User Not Found" alert, preventing the creation of orphaned records in the database.

5.4.2 User Acceptance Testing (UAT)

User Acceptance Testing was conducted with the primary stakeholder, Cikgu Farah. The system was deployed on a laptop connected to the school's barcode scanner.

- **Feedback on OCR:** The librarian noted that while the OCR struggled with cursive handwriting, the split-screen verification interface made it significantly faster to type the data than referring to the physical book manually.
- **Feedback on Scanning:** The Student Assistants found the simplified dashboard easy to use, requiring less than 5 minutes of training to start processing returns.

5.5 Chapter Summary

This chapter described the implementation and testing of *ePustaka Munshi*. The system was successfully built using React and Supabase, with Tesseract.js powering the digitization module. The interfaces were designed to solve specific operational problems, such as data verification and rapid transaction processing. Testing confirmed that the system meets the functional requirements defined in the SRS, and UAT feedback indicated that the system is ready for deployment to assist in the daily operations of SMK Abdullah Munshi's library.

REFERENCES

- Alkhidmah Journal. (2023). Digital Library Management System Study. *Journal of Information Systems and Technology*, 4(2), 45-58.
<https://ejurnalqarnain.stisnq.ac.id/index.php/ALKHIDMAH/article/view/501>
- Amir, S., & Harun, M. (2021). The impact of library automation on school resource management in Malaysia. *Malaysian Journal of Library & Information Science*, 26(1), 1-15.
- CDS Online. (n.d.). Open Source Library Systems & Digitisation. Retrieved October 15, 2025, from <https://cdsol.com.my/web/open-source-library-systems-digitisation/>
- IEEE Computer Society. (2014). IEEE Standard for Software Requirements Specifications (IEEE Std 830-1998). IEEE.
- Libsys Malaysia. (n.d.). Library Management System Solutions. Retrieved November 2, 2025, from <https://www.libsys.my/>
- Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach* (9th ed.). McGraw-Hill Education.
- Rose, L., & Shaukat, M. (2019). Optical Character Recognition (OCR) for digitizing historical documents: Challenges and techniques. *International Journal of Computer Applications*, 178(3), 12-18.
- Schwaber, K., & Sutherland, J. (2020). *The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game*. Scrum.org.

Smith, R. (2021). Tesseract.js Documentation: Pure Javascript OCR for 62 Languages. Retrieved December 10, 2025, from <https://tesseract.projectnaptha.com/>

Sommerville, I. (2016). Software Engineering (10th ed.). Pearson Education Limited.

Supabase. (2024). Supabase Documentation: The Open Source Firebase Alternative. Retrieved November 20, 2025, from <https://supabase.com/docs>

Appendix A SDD DOCUMENT

Appendix B SRS DOCUMENT

Appendix C STD DOCUMENT

Appendix D Other Relevant Documents